Method of Moment - Generating Functions

Theorem 6.1 : (Uniqueness)

If the two random variables, X and Y, have the same moment-generating functions, $m_Y(t) = m_X(t)$ for all t, then X and Y have identical probability distributions.

Method of Moment - Generating Functions

Consider $U = h(Y_1, Y_2, \ldots, Y_n)$, a function of the random variables $Y_1, Y_2, \ldots, Y_n$.

- Find the moment-generating function for U, $m_U(t)$.
- Compare $m_U(t)$ with other known moment-generating functions. If $m_U(t) = m_V(t)$ for all values of t, then U and V have identical distributions.

Example 1:

The joint density of X and Y is given as

$$f(x,y)=\begin{cases}\dfrac{1}{\theta^2} & , \quad 0<y<\theta \text{ and } 0<x<\theta, \\ 0 & , \quad \textit{else}\end{cases}$$

Note: X and Y are independent $Uniform(0,\ \theta)$ random variables.

a) What is the probability distribution of $U=aY+b$?

b) What is the probability distribution of $U = -a\ln(Y),\ a > 0$?

Example 4:

The amount of time waiting to be served, Y, is described by the following density

$$f(y) = \begin{cases} \frac{1}{\theta} e^{-y/\theta} & , \quad y > 0 \\ 0 & , \quad else \end{cases}$$

a) What is the probability distribution of $U = aY + b$, where $a > 0$?

Example 3:

Suppose $Z \sim Normal(0,1)$.

a) What is the probability distribution of $U = aZ + b$, where $a > 0$?

Suppose $Z \sim Normal(0,1)$.

b) What is the probability distribution of $U = Z^2$?

Theorem 6.2: Let $Y_1, Y_2, \ldots, Y_n$ be independent random variables with moment-generating functions $m_{Y_1}(t), m_{Y_2}(t), \ldots, m_{Y_n}(t)$, respectively.

If $U = Y_1 + Y_2 + \cdots + Y_n$ then,

$$m_U(t) = m_{Y_1}(t) \times m_{Y_2}(t) \times \cdots \times m_{Y_n}(t) = \prod_{i=1}^{n} m_{Y_i}(t).$$

Special Case: If $Y_1, Y_2, \ldots, Y_n$ are also identically distributed with moment-generating function $m_Y(t)$ then,

$$m_U(t) = \left[m_Y(t)\right]^n.$$

Example 4:

The amount of time waiting to be served, Y, is described by the following density

$$f(y) = \begin{cases} \frac{1}{\theta} e^{-y/\theta} & , \quad y > 0 \\ 0 & , \quad else \end{cases}$$

Consider n independent waiting times, $Y_1, \ldots, Y_n$, all following the distribution above.

c) What is the probability distribution of the total time, $\sum_{i=1}^{n} Y_i$?

<u>Example 19:</u> Consider independent random variables, $X_i \sim Poisson(\lambda_i),\ i = 1,\ldots,n.$ Find the moment generating function of Y, where $Y = \sum_{i=1}^{n} X_i$, and $X_i \sim Poisson(\lambda_i),\ i = 1,\ldots,n.$ What can you say about the distribution of the random variable Y?